



Propeller Partial Ground and Ceiling Effect Prediction

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Our propeller partial ground-effect study was extended to annulus ground-plates. Correlations between force data and flow visualization indicate that onset of strong outward radial flow on the ground plate, as normalized ground-separation distance decreases, appears together with thrust-augmentation and power-reduction in ground effect. Thrust increase in inverse to ground-effect, or ceiling effect, was found to depend just on ground plate area, regardless of plate geometry, whether the plate is an annulus or a disk. Our earlier infinite-ground correlation was modified for ground/ceiling plate geometries studied here, with an error of less than 6%.

I. Nomenclature

B = Ratio of propeller diameter to circular ground-plate outer diameter; $B = \frac{d_p}{D}$

C_T = Propeller Thrust Coefficient; $C_T = \frac{T}{\rho n^2 D^4}$

C_P = Propeller Power Coefficient; $C_P = 2\pi C_Q = \frac{P}{\rho n^3 D^5}$

C_Q = Propeller Torque Coefficient; $C_Q = \frac{Q}{\rho n^2 D^5}$

D = Propeller Diameter, (m)

d_p = Ground-plate Diameter, (m)

d_o = Ground-plate Opening Diameter, (m)

D_p = Drag on Ground-Plate, (N)

h = Distance to the ground-plate, (m)

n = Propeller rotational speed per second,

O = Ratio of annulus inner diameter to propeller diameter; $O = \frac{d_o}{D}$

P = Power, (W)

P_{def} = Normalized Power Required Deficit;

p = Pressure, (Pa)

γ = Propeller Pitch, (m)

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$$\begin{aligned}
Q &= \text{Torque } (Nm) \\
R &= \text{Propeller Radius } (m) \\
T &= \text{Thrust, } (N) \\
\rho &= \text{Air Density, } (kg/m^3) \\
\sigma &= \text{Propeller Solidity; } \sigma = \frac{S}{0.25\pi D^2}
\end{aligned}$$

Subscripts:

$$\begin{aligned}
OGE &= \text{Out of Ground Effect} \\
IGE &= \text{In Ground Effect} \\
P &= \text{Predicted}
\end{aligned}$$

II. Introduction

Ground effect on lifting rotor/propeller performance has been studied theoretically [1] and experimentally ([2], [4]–[6], [8], [11]) for decades. Placing the propulsor adjacent and parallel to a ground-plane offers two benefits: an increase in thrust-produced, and a decrease in power-required, for a given rotational speed. In earlier work [12], we considered the extreme ground effect, where the ratio of ground plane stand-off distance to propeller diameter is 0.1 or less. We also reversed the propeller, to “ceiling effect”. The present work continues with alternative finite ground-plate geometries.

A) Prior Work

Our previous study [12] assumed an infinite ground-plate ($B = \infty$), where the ratio of ground-plane physical extent to that of the propeller diameter was greater than four. Multiple APC thin electric propellers [17] were tested in the experiment. Normalized power required at constant thrust, P_{IGE}/P_{OGE} , was used to evaluate the performance of the propeller in ground effect. Positive h/D represents ground effect, while negative h/D is ceiling effect. A correlation, applicable to of pitch-to-diameter ratio of $0 < \gamma/D < 0.636$, was proposed:

$$\frac{P_{IGE}}{P_{OGE}} = \begin{cases} 1 - \left(1 - \frac{P_c}{P_{OGE}}\right) * \sigma * 100 / \left(\frac{\gamma}{D}\right)^2 & (0 \leq \frac{h}{D} \leq 0.5) \\ 1 - \left(1 - \frac{P_c}{P_{OGE}}\right) * \sigma * 100 / \left(\frac{\gamma}{D}\right)^1 & (-0.5 \leq \frac{h}{D} \leq 0) \end{cases} \quad (1)$$

where:

$$\frac{P_c}{P_{OGE}} = \begin{cases} -0.0513 \left(\frac{h}{D}\right)^2 + 0.0508 \left(\frac{h}{D}\right) + 0.9873 & (0 \leq \frac{h}{D} \leq 0.5) \\ -0.1449 \left(\frac{h}{D}\right)^2 - 0.1277 \left(\frac{h}{D}\right) + 0.9707 & (-0.5 \leq \frac{h}{D} \leq 0) \end{cases} \quad (2)$$

Next, a finite ground plate was instantiated as a circular disk [13]. The ratio of ground-plate diameter to propeller diameter, $B = \frac{d_p}{D}$, was varied from 0.5 to 1.0; for nominally infinite ground, $B = \infty$. $B = 0.5$ was seen to evince nearly no ground-effect, while $B = 1$ approximates $B = \infty$. [12] and [13] captured notable difference in ground effect with the ratio of propeller pitch to diameter, γ/D ; higher γ/D had attenuated ground effect. This trend was imputed to likely stall of the higher γ/D propellers in hover – a condition exacerbated by the ground plate.

Near-surface flow visualizations via a dense array of tufts, and separately with smoke, are shown in Figure 1 [13], for the APC 17x7 propeller at $h/D = 0.2$ and $B = \infty$. A projected “stagnation streamline”, approximately circular, of diameter on the order of half of the propeller diameter is observed from the tuft flow visualization. Exterior to this circular boundary, the projected streamlines appear to be radially outward. Interior to the circular boundary, the projected flow can be described as being tangential and swirling, without discernible radial component. This phenomenon is also observed in the smoke flow visualization. Based on the force measurement and flow visualization result, it is very likely that the thrust argumentation and power reduction in ground effect relate to trends in the tangential and radial flow region.

The idea of an annulus ground plate then emerges, where the center region of the plate is removed, which eliminates the tangential flow region, as an opposite condition to that of the circular ground plate with $B < 1.0$.

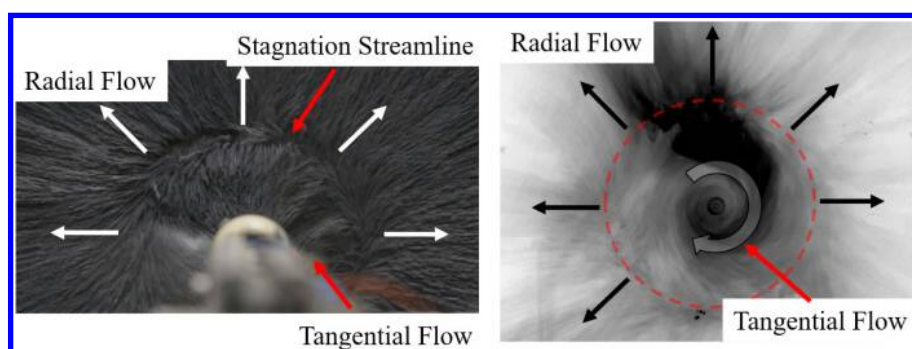


Figure 1. Tuft (left) and smoke (right) flow visualization of near-surface flowfield, $h/D = 0.2$.

III. Experimental Setup

Experiments were conducted on a thrust-stand at the University of Dayton. Propellers were driven by an E-Flite Power 60-400kV brushless out-runner electric motor, driven by a PSW 30-108 constant-voltage power supply. A photodiode was used to measure propeller RPM. A schematic of the setup is given in Figure 2.

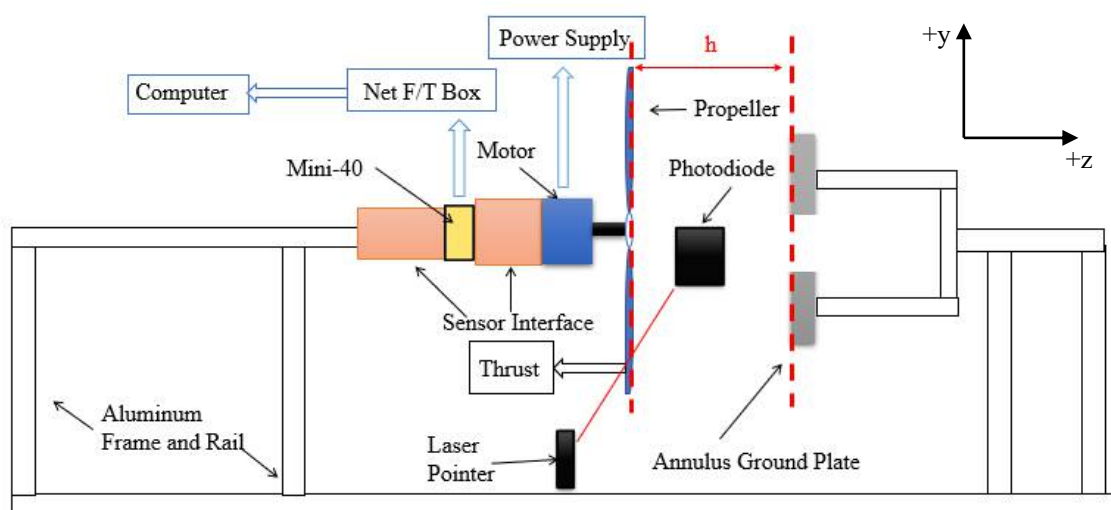


Figure 2. Schematic (side view) of thrust stand and ground-plate

A) Thrust and Torque Measurements

The propeller motor was attached to an ATI Industrial Automation Mini-40 six-component force balance (www.ati-ia.com [15]). Thrust is the axial-force along the z-axis of the sensor, and torque is a rotational moment

about the z-axis. The balance bolts to an aluminum-extrusion frame secured to the lab floor. The balance has a maximum of 120N force on its Z axis and maximum torque of 2 Nm about its Z axis. The encountered peak thrust and torque were 40N and 1.2 Nm, respectively. The ground plate was attached to an ATI Gamma [16] six-component force balance to measure the normal force acting on it.

The data sampling rate was 100 Hz, for a 15-second duration at each data-point, with two runs to improve data repeatability for each data point (value of rpm; propeller pitch/diameter; and normalized ground-standoff distance h/D). In an effort to minimize sensor-drift, tare values are taken before and after each test, and then the average of the two tares is subtracted from the respective thrust and torque readings.

To attenuate blockage due to the thrust stand itself, which is to say, blockage unrelated to ground-effect, the propeller-disk is >1 diameter behind any transverse members of the aluminum frame, and the 3D-printed plastic streamline-cowling covering the sensor is $< 2\%$ of the propeller disk-area. As is usually the case, power-wires from the speed controller to the motor, cross the metric junction.

To achieve negative h/D , the propeller is attached reversely on the motor. The motor rotation direction is also reversed. The APC 17x7 “thin electric” propeller (Table 1) is selected, based on a good performance found in previous study [12,13]. blade Reynolds number is calculated at 70% blade-span, for the peak rpm achieved for each given propeller. Thrust coefficients at 5,400 RPM are taken and then divided by the solidity of the blade to get C_T/σ .

Table 1. Propeller Properties

Propeller	Pitch (γ)/ Diameter (D)	Solidity (σ)	Peak Tip Speed (m/s)	Peak Reynolds Number (x1000)	Power Level C_T/σ
17x7	0.412	0.0846	142.0	190	0.889

B) Ground Plate Nomenclature

As before [13], we define the ratio of the outer diameter of the ground plate and that of the propeller as $B = \frac{d_p}{D}$. The ratio of the ground plate inner diameter to the propeller diameter is $O = \frac{d_o}{D}$. Thus, a circular plate with half of the propeller diameter is $B_{0.5}O_{0.0}$, an annulus with the same outer diameter as the propeller and an opening half of the propeller diameter is $B_{1.0}O_{0.5}$, an infinite ground plate with an opening half of the propeller diameter is $B_{\infty}O_{0.5}$ (Figure 3), and the usual infinite ground-plate becomes $B_{1.0}O_{0.0}$. Table 2 lists the test conditions.

Table 2. Ground Plate Properties

Infinite Ground ($B = \infty, O = 0.0$)	Circular Ground ($B \neq 0, O = 0.0$)	Annulus Ground ($B = 1.0, O \neq 0.0$)	Annulus Infinite Ground ($B = \infty, O \neq 0.0$)
$B_{\infty}O_{0.0}$	$B_{0.5}O_{0.0}$ $B_{0.625}O_{0.0}$ $B_{0.75}O_{0.0}$ $B_{1.0}O_{0.0}$	$B_{1.0}O_{0.5}$ $B_{1.0}O_{0.625}$ $B_{1.0}O_{0.75}$	$B_{\infty}O_{0.625}$ $B_{\infty}O_{0.75}$

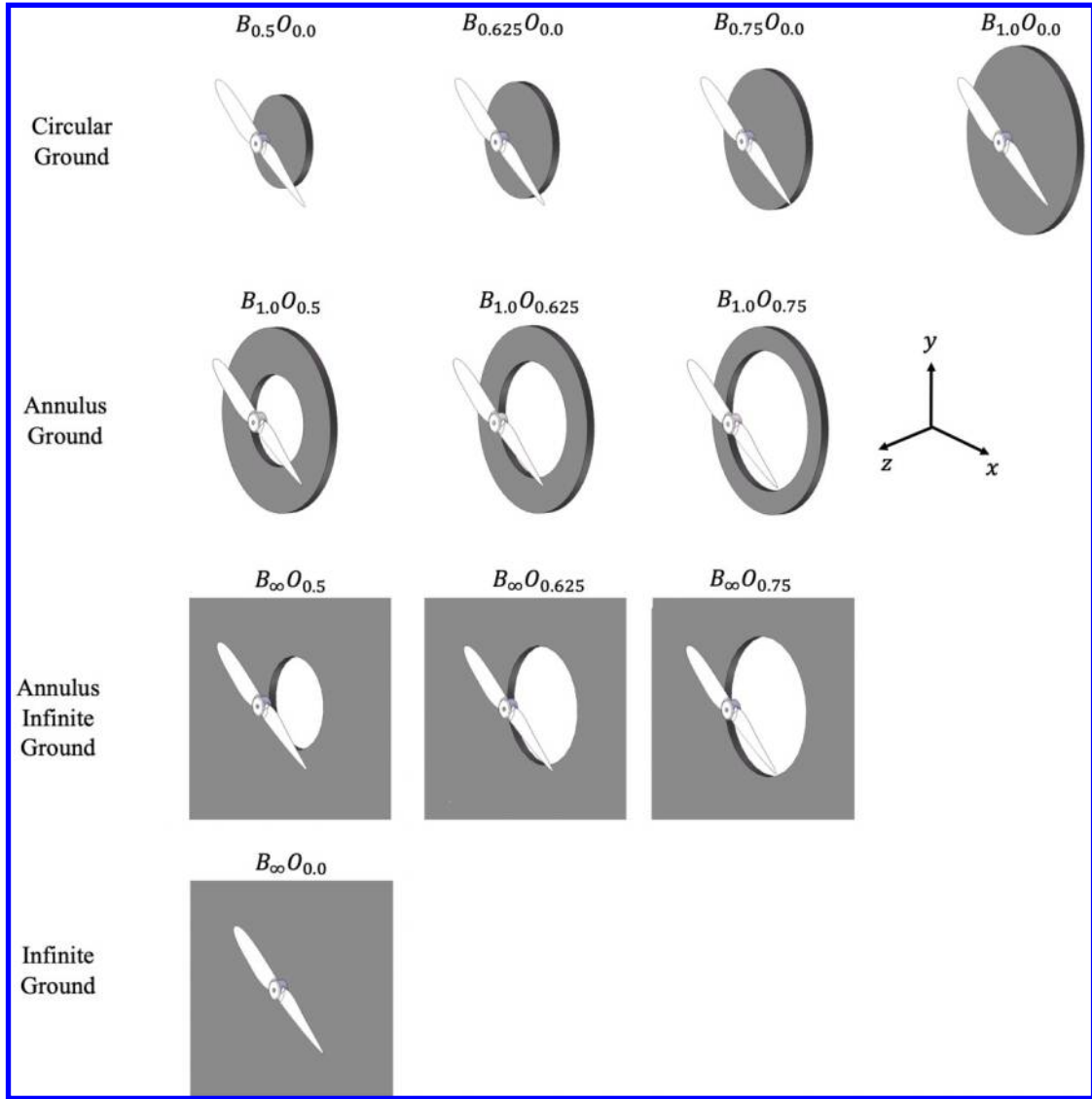


Figure 3. Ground plate naming-convention

C) Flow Visualization

Smoke visualization was performed at RPM ~ 900 . The smoke wand has a diameter of 0.5 inch, and was placed $0.5D$ upstream the propeller disk. A 200-mW laser was used as a light source.

D) Unsteady Pressure Transducer Measurement

Seven Kulite XTEL-10L-190 pressure transducers (www.kulite.com) [18] were placed flush on the circular ground plate and annulus ground plate, shown in Figure 4. The pressure transducer is located at various radial locations shown in Table All have a range of ± 170 Kpa gage pressure. Data was collected at a sampling rate of 5kHz for 5 seconds. The time-averaged pressure is then normalized by the spatial average plate pressure calculated from the thrust measurement: $p = 4D_p / \pi d_p^2$ for the $B_{1.0}O_{0.0}$ case at the respective h/D . Pressure data were taken in the h/D range from 0.1 to 0.5, in 0.1 h/D increments.

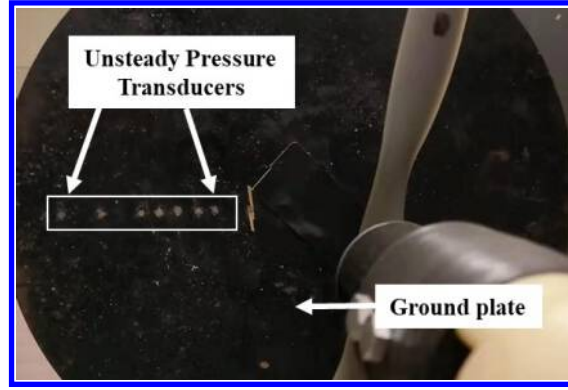


Figure 4. Placement of the Kulite pressure transducers on the ground plate

Table 3. Pressure Transducer Placement

Ground Plate	Pressure Transducer Radial Location (r/R)
$B_{1.0}O_{0.0}$	0.24, 0.35, 0.47, 0.59, 0.71, 0.82, 0.94
$B_{0.5}O_{0.0}$	0.24, 0.29, 0.35, 0.41, 0.47
$B_{0.625}O_{0.0}$	0.24, 0.29, 0.35, 0.41, 0.47, 0.53, 0.59
$B_{0.75}O_{0.0}$	0.24, 0.35, 0.41, 0.47, 0.53, 0.59, 0.71
$B_{1.0}O_{0.5}$	0.53, 0.59, 0.65, 0.71, 0.76, 0.82, 0.88
$B_{1.0}O_{0.625}$	0.65, 0.71, 0.76, 0.82, 0.88, 0.94
$B_{1.0}O_{0.75}$	0.76, 0.82, 0.88, 0.94

IV. Results

A. Thrust and Power Data

The normalized power required at constant thrust (P_{IGE}/P_{OGE}) for different annuli is shown in Figure 5, along with $B_{1.0}O_{0.0}$ [12]. For the ceiling effect, where $h/D < 0$, the result for all annuli follows the general trend, where P_{IGE}/P_{OGE} decreases with decrease in h/D . $B_{1.0}O_{0.5}$, which has the smallest opening among all cases, results in a larger reduction in P_{IGE}/P_{OGE} ; approximately 80%, when compared to $B_{1.0}O_{0.0}$. $B_{1.0}O_{0.75}$ and $B_{\infty}O_{0.75}$, which have the largest opening, result in the smallest reduction in P_{IGE}/P_{OGE} . The difference between these two cases is negligible, with $B_{\infty}O_{0.75}$ having a slightly greater drop in P_{IGE}/P_{OGE} than $B_{1.0}O_{0.75}$, once again confirming that $B_{1.0}O_{0.0}$ duplicates the results of the infinite ground/ceiling. $B_{1.0}O_{0.625}$ and $B_{\infty}O_{0.625}$ fall between $B_{1.0}O_{0.5}$ and $B_{1.0}O_{0.75}$, showing a decreasing ceiling effect with increase in annular opening. Again, the difference between these two cases is minor.

For $h/D > 0$, significant deviations are seen between the different cases. $B_{1.0}O_{0.5}$ still follows the general trend in ground effect, where P_{IGE}/P_{OGE} reduces with decrease in h/D . However, $B_{1.0}O_{0.75}$ and $B_{\infty}O_{0.75}$ both show a small increment in P_{IGE}/P_{OGE} with the decrement of h/D . $B_{1.0}O_{0.75}$ shows a significant increment in P_{IGE}/P_{OGE} while $B_{\infty}O_{0.75}$ case has a small drop. Propeller momentum theory [14] states that the area of the

propeller far wake is half of the propeller disk area. This corresponds to $B = 0.707$, which is smaller than the prescribed 0.75. This may explain attenuation of ground effect trends for $B_{1.0}O_{0.75}$ and $B_{\infty}O_{0.75}$.

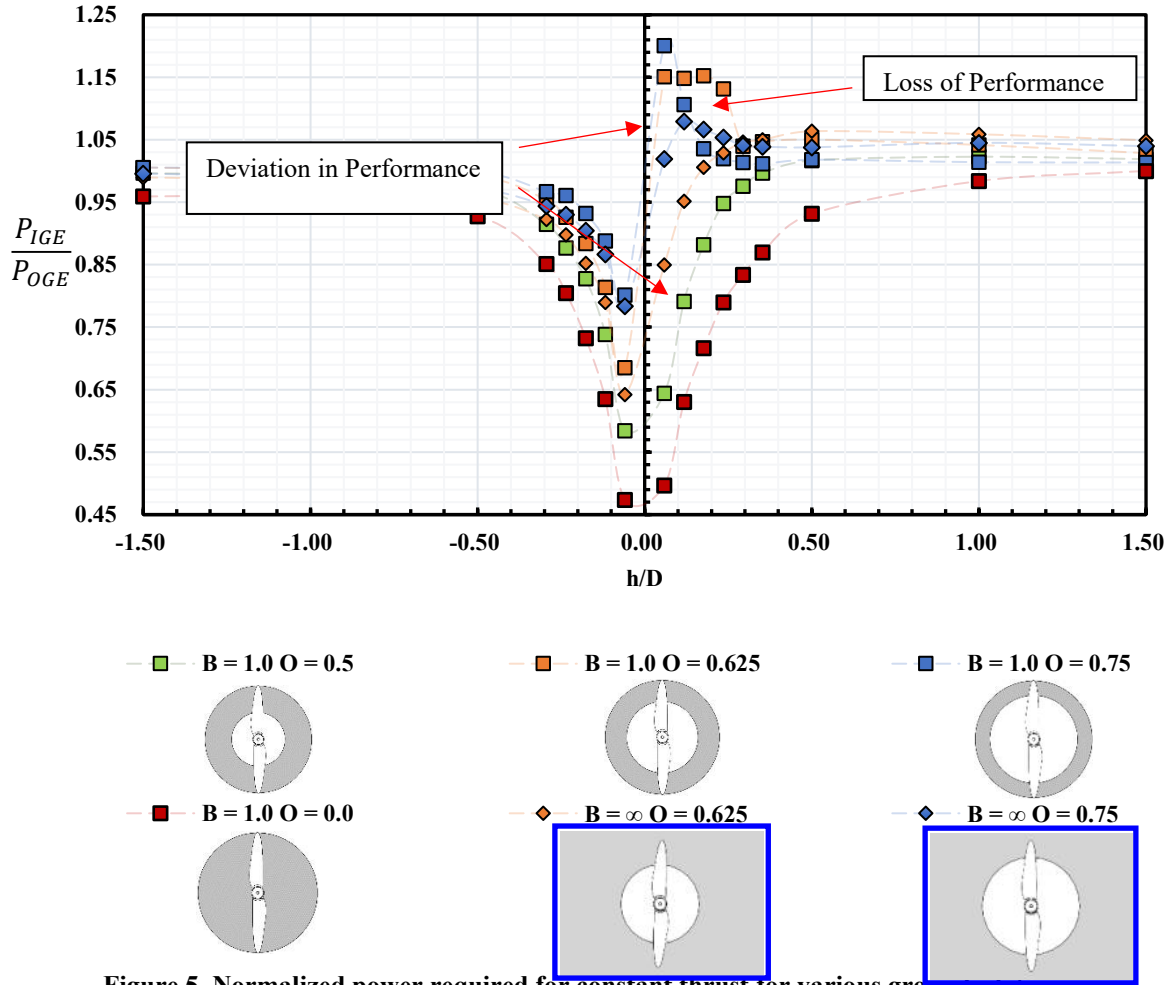


Figure 5. Normalized power required for constant thrust for various ground-plate cases

Unlike the ceiling effect, $B_{1.0}O_{0.625}$ and $B_{\infty}O_{0.625}$ deviate significantly, especially below $h/D < 0.3$. While $B_{\infty}O_{0.625}$ follows the trend seen in traditional ground effect, $B_{1.0}O_{0.625}$ has a sudden increment in P_{IGE}/P_{OGE} at $h/D < 0.3$, similar to $B_{1.0}O_{0.75}$.

The aforementioned trends were examined by plotting the individual thrust and power coefficients, normalized by their OGE reference values in Figure 6 and Figure 7, respectively. Recall that the result for all annuli at $h/D < 0$ (ceiling effect) follows the expected trends, with $B_{1.0}O_{0.5}$ resulting in the largest increment in thrust coefficient, while $B_{1.0}O_{0.75}$ has the smallest increment.

For $h/D > 0$, the IGE increment for the $B_{1.0}O_{0.5}$ case, in thrust follows closely to the $B_{1.0}O_{0.0}$. Recall from the surface tuft flow visualization result in Figure 1, the diameter of the “stagnation streamline” for 17x7 propeller is approximately 50% of the propeller disk. $B_{1.0}O_{0.5}$ case eliminates the region of inward tangential flow while remains most of the outward tangential flow region. This indicates that the changes in the thrust coefficient is a result of outward radial flow, or the deflection of the propeller wake in the radial direction.

Except for smaller h/D values, changes in the thrust coefficient for both $O_{0.75}$ cases is negligible, with a small deviation at small h/D value, similar to the trend in **Figure 5**. Again, both $O_{0.625}$ cases have a significant deviation on thrust coefficient, with $B_{\infty}O_{0.625}$ following the general trend in ground effect with an increment in

thrust coefficient at small h/D . While the $B_{1.0}O_{0.625}$ case behaves more likely to the $B_{1.0}O_{0.75}$ case with a significant drop in the thrust coefficient.

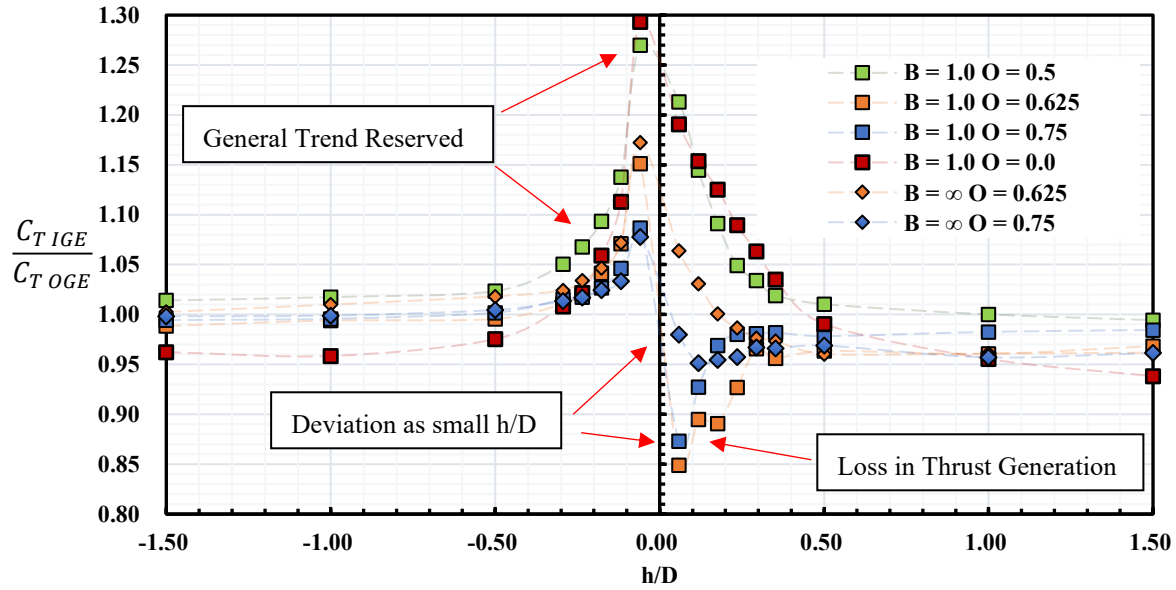


Figure 6. Normalized thrust coefficient of cases in Figure 5

The changes in the power coefficient for the ceiling effect ($h/D > 0$) follows the ground effect trend with a decrement of power coefficient at small h/D value. For ground effect result, the $B_{1.0}O_{0.5}$ case has a minor decrement in power coefficient with the decrement in h/D , however, the magnitude is much smaller when compared to the $B_{1.0}O_{0.0}$ case. This indicates that the difference between these two cases in power required at constant thrust (Figure 5) is due to the differences seen in the power coefficient. The $B_{\infty}O_{0.625}$, $B_{1.0}O_{0.75}$ and $B_{\infty}O_{0.75}$ cases has almost no changes in the power coefficient while the $B_{1.0}O_{0.625}$ has a drop in power coefficient at very small h/D , possibly due to the reduction in the thrust coefficient.

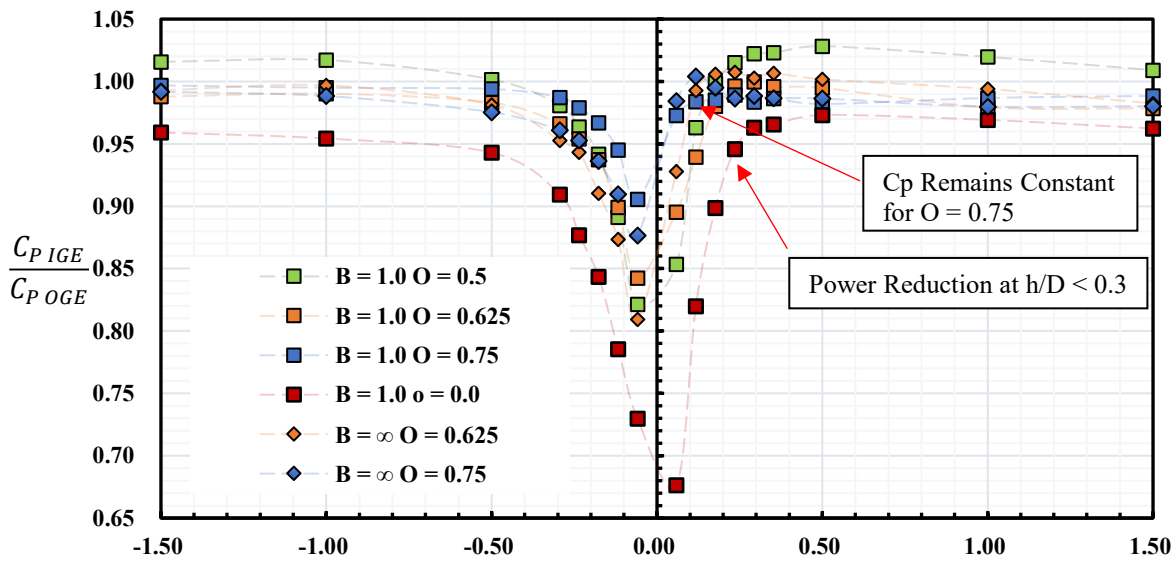


Figure 7. Normalized power coefficient for the cases in Figure 5

B. Flow Visualization

Figure 8 shows smoke injected near the plate at $\sim 50\%$ propeller radius, for $B_{1.0}O_{0.5}$, $B_{1.0}O_{0.625}$, $B_{\infty}O_{0.625}$ and $B_{1.0}O_{0.75}$ at $0.1 h/D$. There is an inner tangential flow region, and an outer radial flow region, separated by a “stagnation streamline”, noted as a yellow dot in Figure 8. Both types of flow occur in $B_{1.0}O_{0.5}$, with a very strong outward radial flow and inward tangential flow towards the opening on the plate, resembling the infinite-ground case [13]. Again, $B_{1.0}O_{0.5}$ still reflects the general benefits of the ground effect in the force-based experiment: an increment in thrust coefficient and a decrement in power required.

For $B_{1.0}O_{0.625}$ and $B_{1.0}O_{0.625}$, the radial flow region area reduces significantly, while the tangential flow region disappears, leaving the stagnation point at the edge of the annulus opening. This results in less increment in thrust coefficient shown in Figure 6, along with less decrement in power coefficient shown in Figure 7. When we further increase the opening, to $B_{1.0}O_{0.75}$, the majority of the flow go through the opening due to the shrinkage of the propeller wake. Thus, the effect of the annulus is negligible. This agrees with the force data, where the changes in the thrust and power coefficient is negligible for both $B_{1.0}O_{0.75}$ and $B_{\infty}O_{0.75}$.

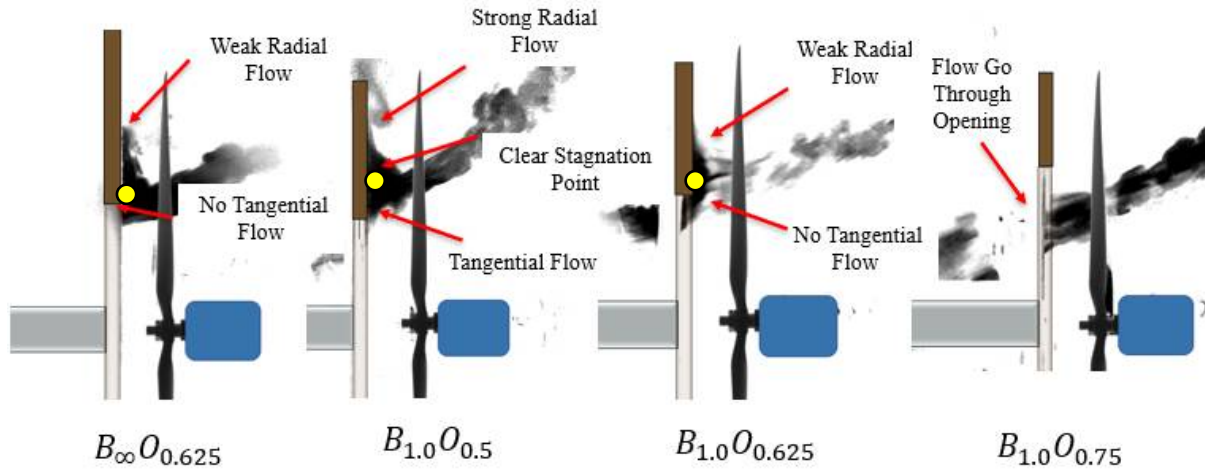


Figure 8. Smoke injection near $0.5D$, for $B_{1.0}O_{0.5}$, $B_{1.0}O_{0.625}$, $B_{\infty}O_{0.625}$ and $B_{1.0}O_{0.75}$ at $h/D = 0.1$.

Smoke visualization around the blade tip for aforementioned the cases is shown below in Figure 9. Strong radial flow is found for both $B_{1.0}O_{0.5}$ and $B_{\infty}O_{0.625}$ cases. Recall from the force/torque measurement result in Figure 6 and Figure 7, both increment in thrust coefficient and reduction in power coefficient is found for both $B_{1.0}O_{0.5}$ and $B_{\infty}O_{0.625}$. Along with the result in Figure 8, evidently the radial flow has a strong relationship with thrust-augmentation and power-reduction in ground effect.

On the other hand, a clear fountain vortex occurs at the tip of the propeller for both $B_{1.0}O_{0.625}$ and $B_{1.0}O_{0.75}$. As the fountain vortex has been shown to increase the induced flow velocity and to reduce the propeller thrust coefficient [17], this results in the loss of thrust coefficient for $B_{1.0}O_{0.625}$ and $B_{1.0}O_{0.75}$, which agrees with the thrust coefficient measurement in Figure 6.

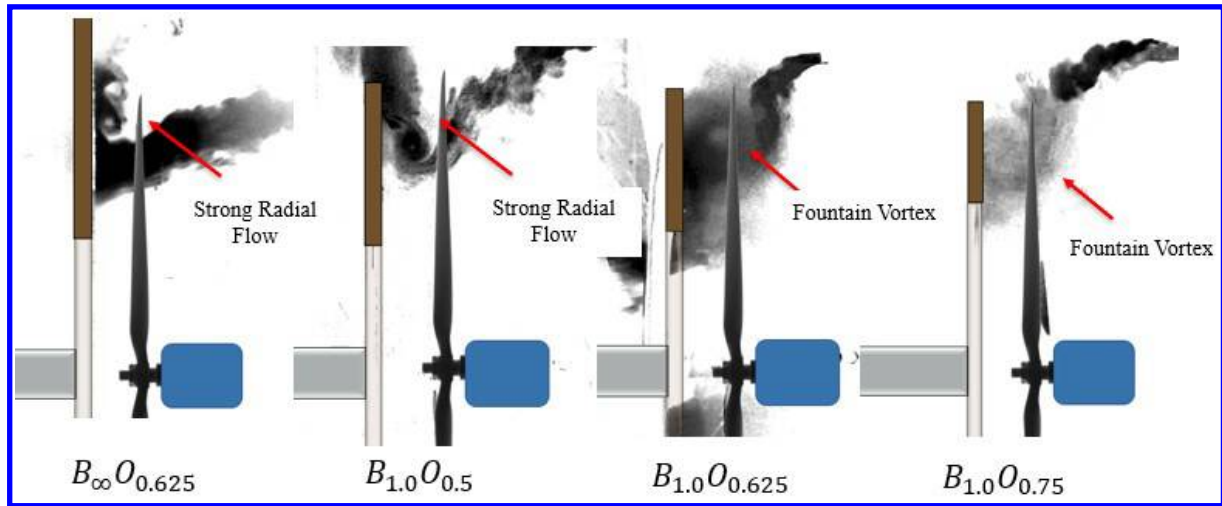


Figure 9. Smoke injection at $\sim 1.0D$, for $B_{1.0}O_{0.5}$, $B_{1.0}O_{0.625}$, $B_{\infty}O_{0.625}$ and $B_{1.0}O_{0.75}$.

C. Synthesis

To what extent do the individual ground-plane-elements' P_{IGE}/P_{OGE} results linearly superimpose? Consider a normalized power-deficit:

$$P_{def} = 1 - \frac{P_{IGE}}{P_{OGE}} \quad (3)$$

P_{def} is calculated for all individual cases, and is then added for $B_{0.75}O_{0.0}$ and $B_{\infty}O_{0.75}$ to represent the P_{def} of an infinite ground plane, and for $B_{0.75}O_{0.0}$ and $B_{1.0}O_{0.75}$ to represent the P_{def} of a $B_{1.0}O_{0.0}$ circular plate. The new value of P_{IGE}/P_{OGE} is then calculated for each case using the summed P_{def} . An example of the summation is given in Figure 10.

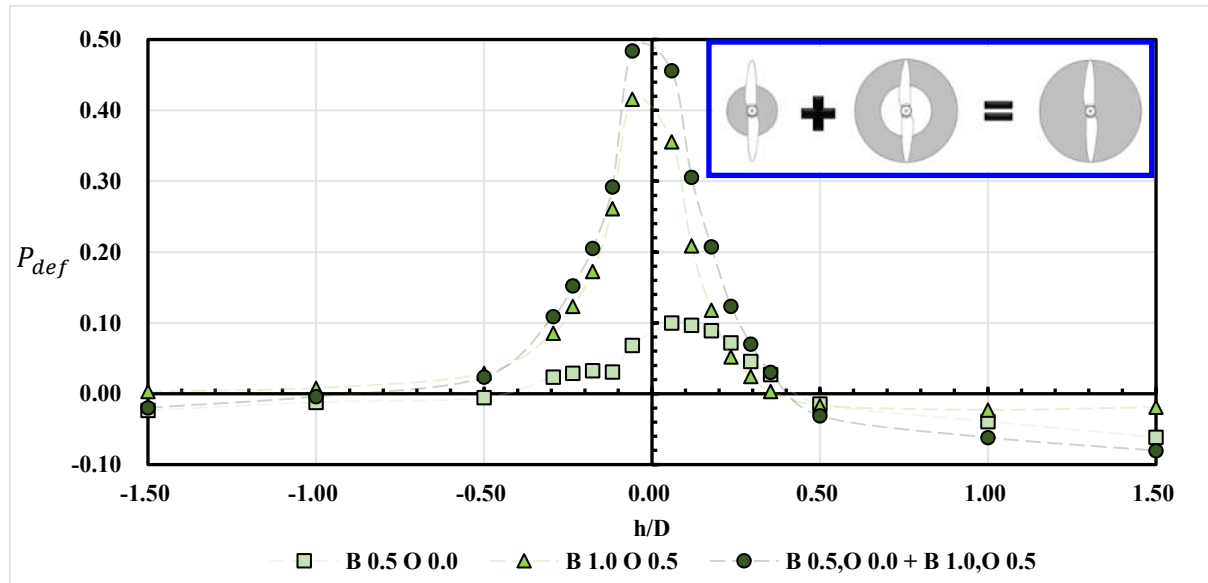


Figure 10. Example of power required deficit summation

This process is then repeated for the rest of the individual cases, which sum to either an infinite ground plane or a $B_{1.0}O_{0.0}$ circular plate. A new power required at constant thrust is then calculated by the summed-up power required deficit, along with the $B_{1.0}O_{0.0}$ case, are shown in Figure 11.

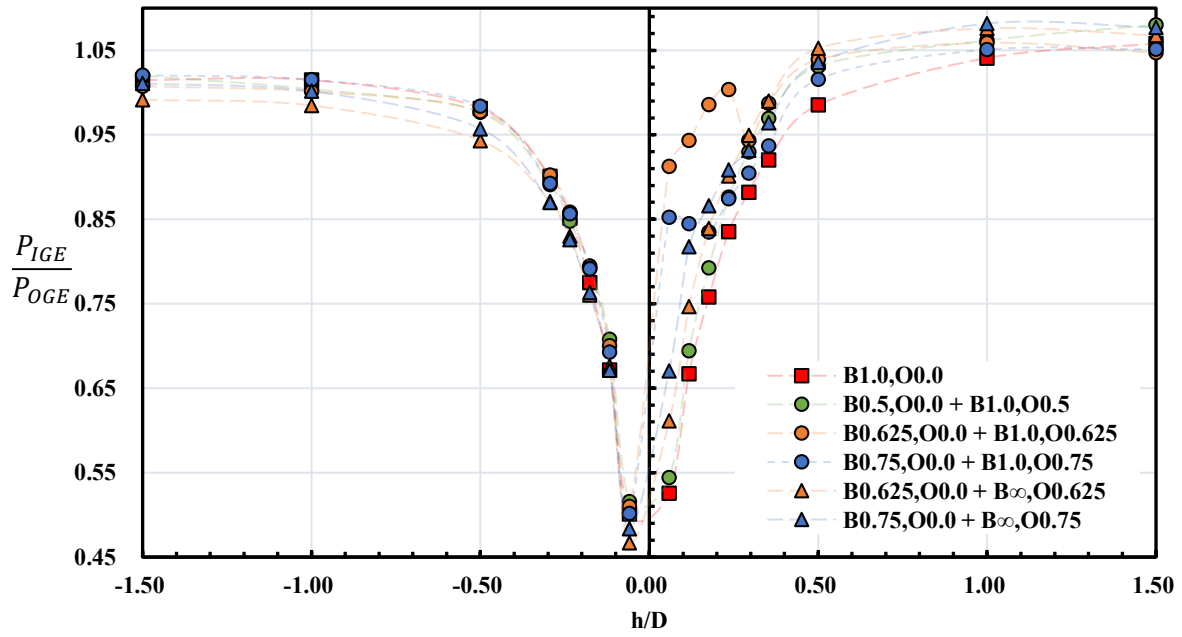


Figure 11. Power required at constant thrust, summing complementary plate-cases

Surprisingly, all curves overlap for the ceiling effect ($h/D < 0$). Recall that the results in last section for the ceiling effect all follow the same trend as the “infinite” ground plane. It is appealing to speculate that the ceiling effect can be represented by a function of the ground plate disk area for any type of plate geometry.

For the ground effect result ($h/D > 0$), only the combination of $B_{0.5}O_{0.0}$ and $B_{1.0}O_{0.5}$ resembles $B_{1.0}O_{0.0}$. $B_{0.625}O_{0.0} + B_{\infty}O_{0.625}$ and $B_{0.75}O_{0.0} + B_{\infty}O_{0.75}$ have a similar trend as infinite-plate ground effect, with a difference of 20%~30% when compared to $B_{1.0}O_{0.0}$. This is caused by the lack of increment in thrust and decrement in power required for the annulus ground condition, as seen in Figure 6 and Figure 7. A deviation occurs for $B_{0.625}O_{0.0} + B_{1.0}O_{0.625}$ and $B_{0.75}O_{0.0} + B_{1.0}O_{0.75}$ around $h/D = 0.3$, which is a result of the abnormal behavior seen in Figure 5. This brings difficulties in predicting the ground effect for annuli.

D. Pressure Measurement.

Pressure transducer data for $h/D = 0.2$ is shown in Figure 12. Recalling that $B_{1.0}O_{0.0}$ resembles the result for an infinite ground plane [13], $B_{1.0}O_{0.0}$ is considered as the baseline, providing a finite area with which to normalize pressure. For the circular plates, the pressure reading is lower than for the baseline case, with a significant drop at the edge of the plate. For $B_{1.0}O_{0.5}$ and $B_{1.0}O_{0.625}$, the pressure distribution resembles that of $B_{1.0}O_{0.0}$ at $r/R > 0.75$, likely as a result of the strong radial flow. For $B_{1.0}O_{0.75}$, due to the fountain vortex seen in Figure 9, the pressure is lower than for $B_{1.0}O_{0.0}$.

At smaller r/R , a relatively lower pressure distribution is found for both $B_{1.0}O_{0.5}$ and $B_{1.0}O_{0.625}$, due to the occurrence of the inward radial flow which replaces the tangential flow as seen in Figure 1. Both phenomenon lead to the result shown in Figure 11, where the partial ground effect result for the circular plate and annulus plate cannot be added together to resemble the performance of a traditional ground effect.

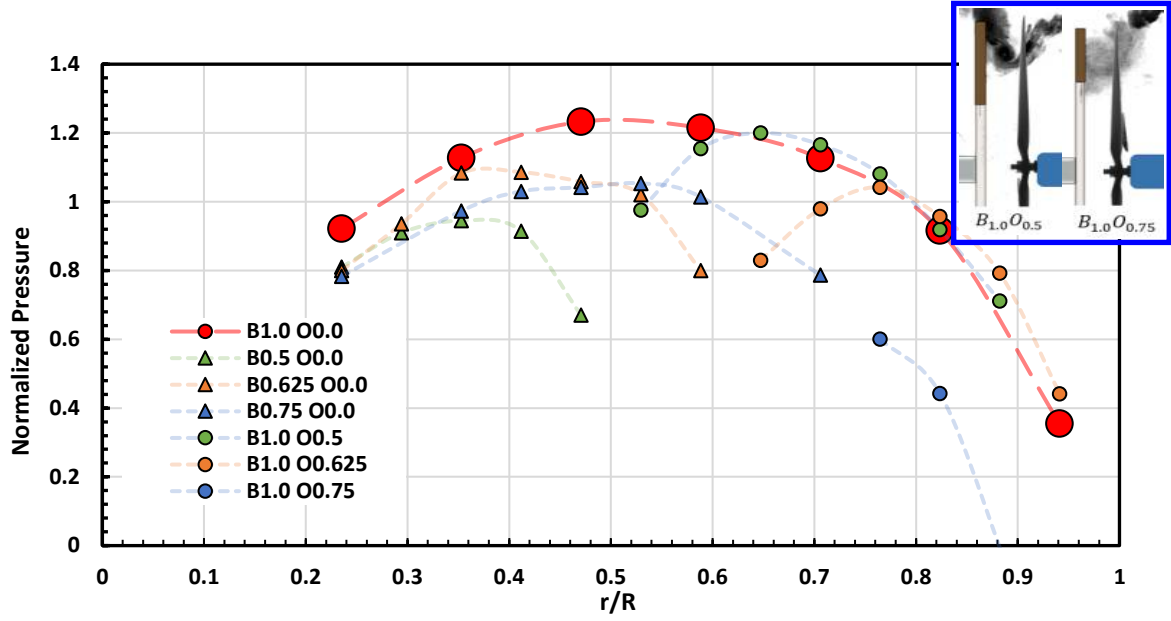


Figure 12. Normalized pressure vs. propeller radial station at $h/D = 0.2$.

The ceiling effect result is shown in Figure 13. Overlap of normalized pressure, whether for circular plates or annuli, is even tighter than in the ground-effect case. This explains the phenomenon in Figure 11, where the thrust in ceiling effect is addable based on plate area.

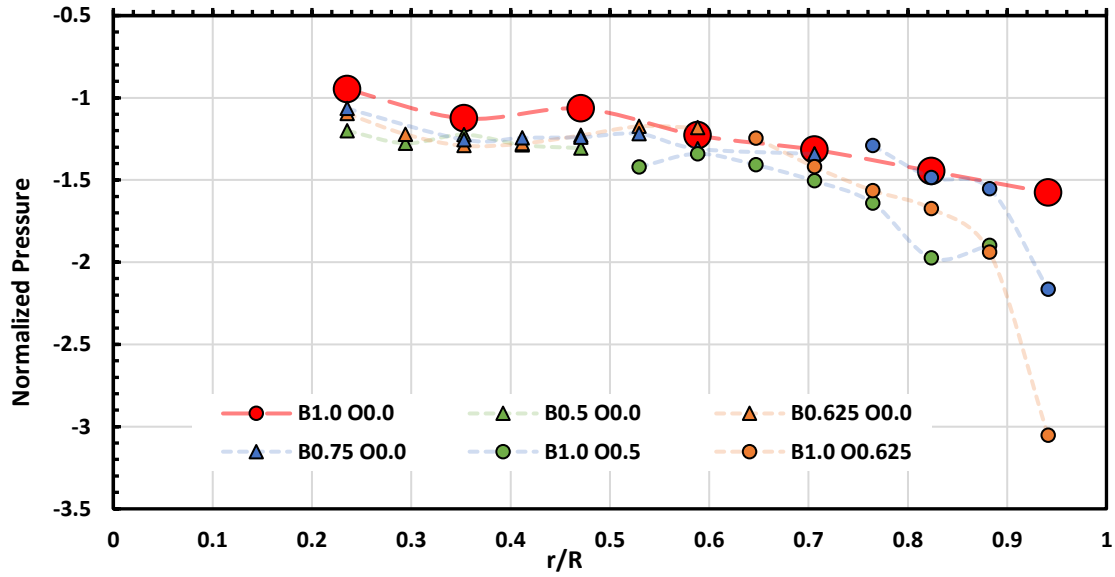


Figure 13. Normalized pressure reading vs. propeller radial station for APC 17x7 at $-0.2 h/D$

V. Correlations.

A. Thrust-based correlations for circular ground plate

The power-required deficit P_{def} for just the circular plates is plotted in Figure 14 as a function of diameter ratio B , for different h/D , with quadratic fits. P_{def} drops to zero around $B = 0.4$, suggesting that ground effect falls to zero with $B < 0.4$. The effective diameter ratio, B_{eff} is introduced as:

$$B_{eff} = \sqrt{(B^2 - 0.4^2)/(1 - 0.4^2)} \quad (4)$$

We now modify Equation 1 to account for different circular plate diameter ratio as:

$$\frac{P_{B\ IGE}}{P_{OGE}} = \begin{cases} 1 - (1 - P_{def}) * B_{eff}^1 & (0 \leq \frac{h}{D} \leq 0.5) \\ 1 - (1 - P_{def}) * B_{eff}^{\sqrt{2}} & (-0.5 \leq \frac{h}{D} \leq 0) \end{cases} \quad (5)$$

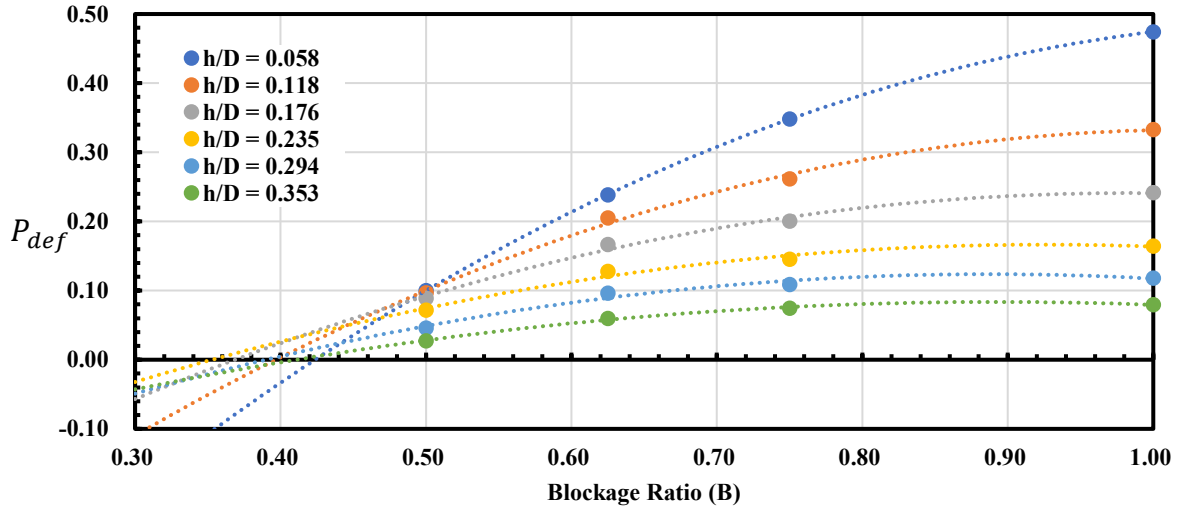


Figure 14. Power-required deficit vs. circular plate diameter ratio at various h/D .

The predicted curve ($P_{B\ IGE}/P_{OGE}$), along with the measured values, is plotted in Figure 15. Scatter in both ground and ceiling effect is small, except for $B = 0.5$ at small h/D , where the fit overpredicts the ceiling effect.

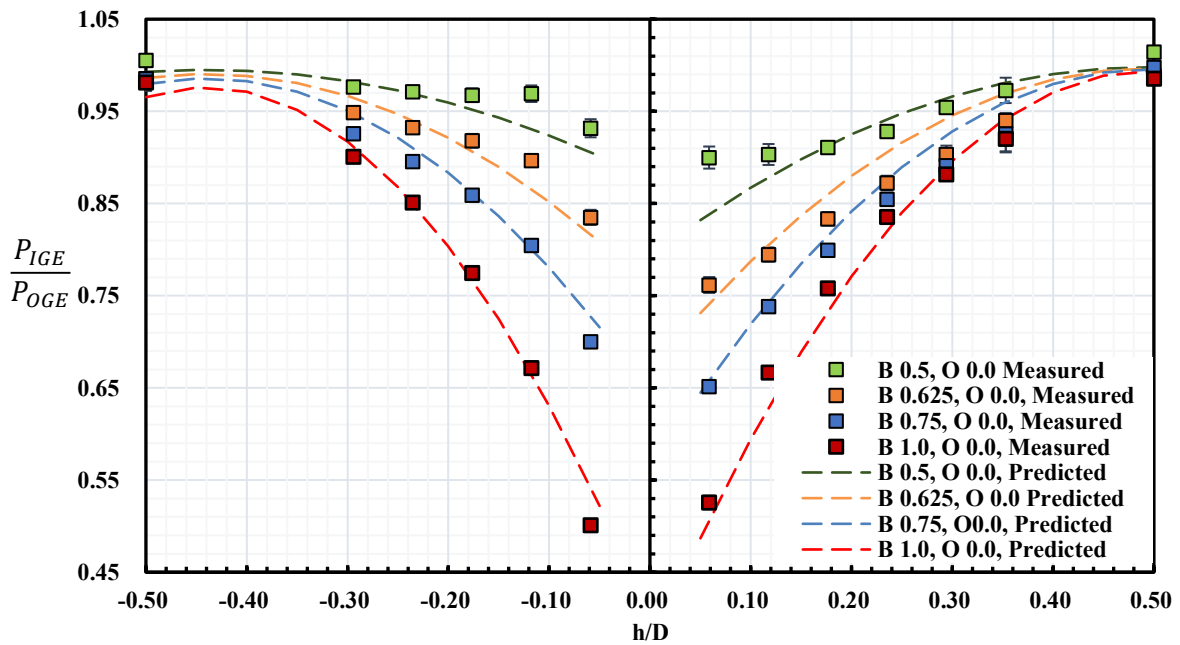


Figure 15. Comparison of predicted changes in $P_{B\ IGE}/P_{OGE}$ and experimental result

B. Data-fit from Pressure Transducers

Since the above method is developed specific for circular ground plate which cannot predict the propeller performance in annulus partial ground or ceiling effect, the second method is developed to fill the gap. Recall from the conclusion of Figure 11 and Figure 13, the ceiling effect can be predicted as a function of plate area. Based on the pressure measurement in Figure 13, the pressure distribution on the ground plate in ceiling effect can be written as a function of r/R . We can then get an expression using a third order polynomial fit.

$$p(r/R) = y = -0.7577 \left(\frac{r}{R}\right)^3 + 0.7716 \left(\frac{r}{R}\right)^2 - 0.8181 \left(\frac{r}{R}\right) - 0.9323 \quad (6)$$

Then, the power required deficit at partial ceiling effect can be written as:

$$P_{\text{def Ce}} = \frac{\int_0^B P(r/R) * 2\pi r dr}{\int_0^1 P(r/R) * 2\pi r dr} * P_{\text{def}(B1.0 \ 00.0)} \quad (7)$$

Thus, the predicted power required at constant thrust for annulus ceiling effect can be written as:

$$\frac{P_{\text{Ce IGE}}}{P_{\text{OGE}}} = 1 - P_{\text{def C}} = 1 - \frac{\int_0^B P\left(\frac{r}{R}\right) * 2\pi r dr}{\int_0^1 P\left(\frac{r}{R}\right) * 2\pi r dr} * P_{\text{def}(B1.0 \ 00.0)} \quad (8)$$

This is plotted in Figure 16, using experimental data from Figure 5. The model slightly overpredicts in the $B_{1.0}O_{0.625}$ and $B_{1.0}O_{0.75}$ case.

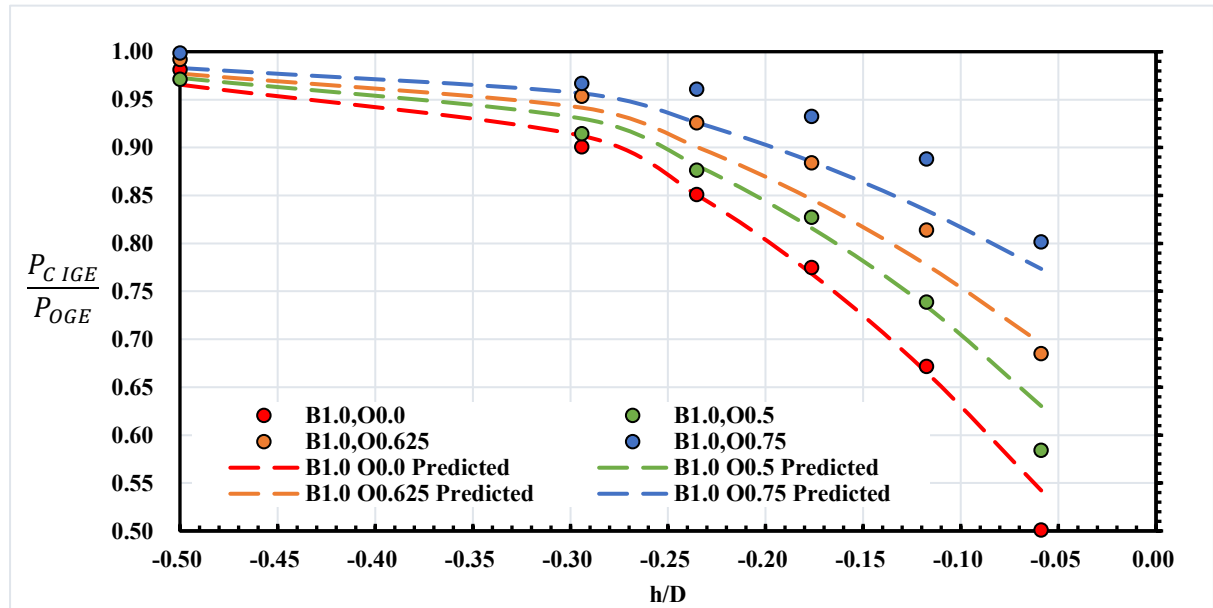


Figure 16. Comparison of predicted and measured $P_{\text{Ce IGE}}/P_{\text{OGE}}$.

To simplify the prediction model by assuming the pressure distribution of the blockage plate remains constant in ceiling effect, Equation 8 becomes:

$$\frac{P_{\text{Ce IGE}}}{P_{\text{OGE}}} = 1 - \frac{\int_0^B 2\pi r dr}{\pi r^2} * P_{\text{def}(B1.0 \ 00.0)} \quad (9)$$

We can now apply Equation 9 to both circular and annulus cases (Figure 17). The prediction by Equation 9 on the circular plate ceiling effect is as good as Equation 5 shown in Figure 15, while the prediction on the annulus ceiling effect seems to be better than Equation 8. This indicates that the assumption made was reasonable and Equation 9 is suitable for predicting ceiling effect with any type of ground plate.

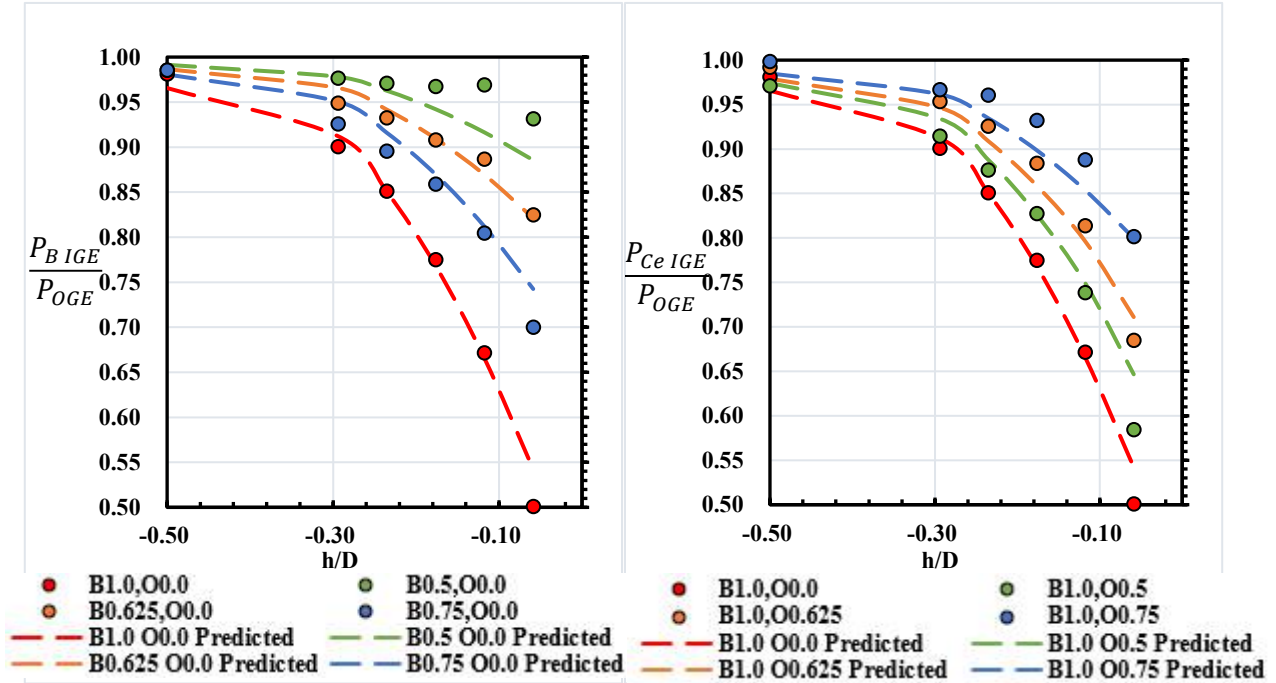


Figure 17. Comparison of predicted changes in circular (left), annulus (right) plates.

VI. Conclusions

Annulus-shaped ground plates were tested, as a supplement to circular plates in ground-effect and ceiling-effect. With small inner diameter of the annulus, strong outward radial was found. results in similar changes in the thrust coefficient when compared to infinite-plate. With greater inner diameter of the annulus, the benefit of the ground effect reduces. Depending on inner diameter, significant deviations were found results for an annulus and an infinite ground plate with same hole-diameter. This is attributed to a “fountain vortex”, which also lead to a reduction of the radial flow strength.

Ceiling effect is found to be additive, in the sense that ground-effect results from a circular disk and an annulus of the same inner-diameter as the disk, matches that of a plane disk, of outer diameter same as the annulus. Likewise, a circular disk and an infinite ground plate, with hole having the same diameter as the disk, sum up to just a simple infinite plate. with a linear combination of partial ground effect results in a ground plate area equals to the propeller disk area, resemble the benefit of an infinite ground plate. This means that ceiling effect depends just on ground plate area, and not on its shape. .

Two correlations were developed. The first model is based on thrust and power measurement, while the second model is based on surface pressure distributions. Both have been shown to have good correlation with trends in normalized power-required in ground effect and ceiling effect.

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